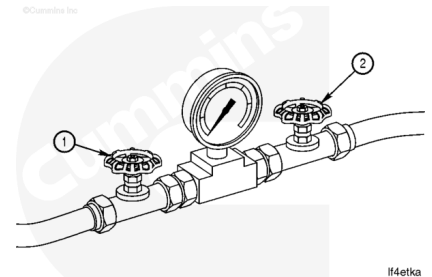


Trainings ▾

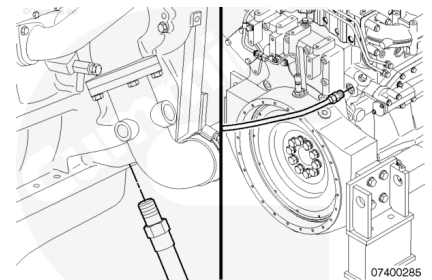
Pressure Differential Test

Use a differential pressure gauge or one pressure gauge with two oil hoses and two valves to eliminate gauge error. Use a gauge with a minimum pressure capacity of 517 kPa [75 psi].

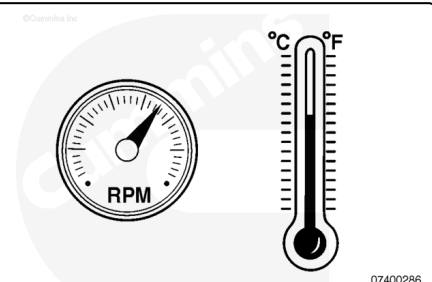


Install a hose into the oil port located on the exhaust side of the engine, under the water pump inlet. This port indicates oil pressure entering the oil coolers.

Install a hose in the oil port located on the right rear of the engine. See illustration for detail. This port indicates oil pressure leaving the oil coolers.



Operate the engine at rated rpm until the oil temperature is at least 112°C [234°F].



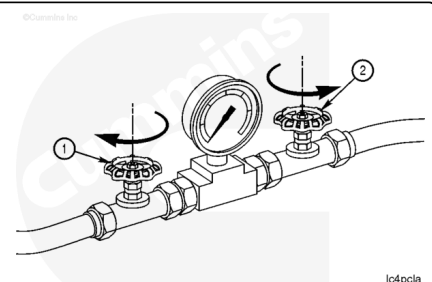
Close valve number 1 and open valve number 2.

Read the oil pressure after the oil has passed through the oil coolers.

Open valve number 1 and close valve number 2.

Read the pressure of the oil being supplied to the oil coolers.

Compare the difference between the two pressures. If the difference is more than the maximum limit specified below, replace the oil cooler elements.



Lubricating Oil Cooler Differential				
	kpa		psi	
Industrial Applications (2100 rpm)		352	MAX	51
Power-Generation (1500 rpm)		248	MAX	36
Power-Generation (1800 rpm)		283	MAX	41

Preparatory Steps

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil.

WARNING

Use engine oil must be disposed of in accordance with local environmental regulations.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

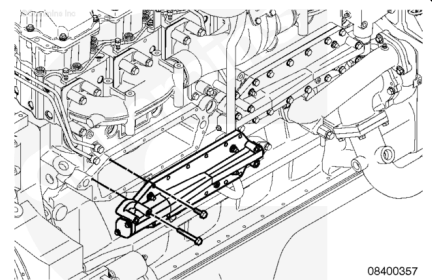
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html>) in Section 8.
- Remove the water pump supply tube, if necessary. Refer to Procedure 008-062 (</qs3/pubsys2/xml/en/procedures/89/89-008-062-tr.html>) in Section 8.
- Remove the turbocharger oil supply and drain lines, if necessary. Refer to Procedure 010-033 (</qs3/pubsys2/xml/en/procedures/89/89-010-033.html>) in Section 10.

Remove

Remove the oil cooler cover and element assembly.

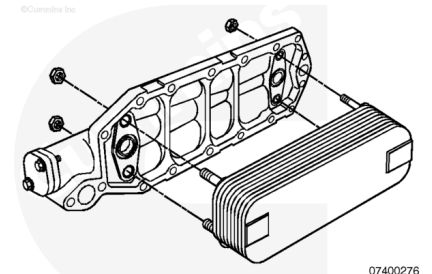
Discard the old gasket.



Disassemble

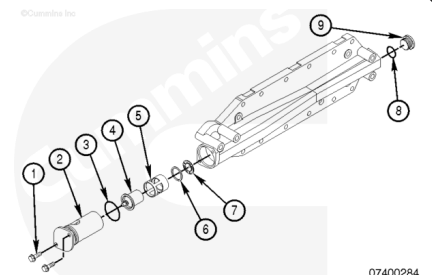
Remove the four nuts and the oil cooler element from each oil cooler assembly.

Remove and discard the six o-rings and two gaskets for each oil cooler assembly.



Remove the following components from the oil cooler cover:

- Capscrews (1)
- Cover (2)
- O-ring (3)
- Thermostat (4)
- Seal housing (5)
- Thermostat seal (6)



- Disk valve (7)
- O-ring (8)
- Plug (9).

Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

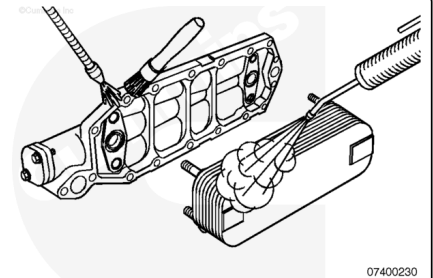
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

Use a solvent that is not harmful to copper.



Replace the oil cooler elements, engine oil, and lubricating oil filters, if any debris is found or the engine has a debris-causing malfunction.

Remove gasket material from the oil cooler cover oil cooler element, and engine block mating surface.

Use steam or hot, high-pressure water. Clean the exterior of the element. Flush the interior of the element until all engine oil is removed.

Dry the element with compressed air.

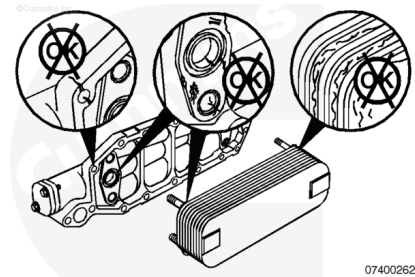
Use solvent to clean the oil cooler cover.

Dry the oil cooler cover with compressed air.

Check the cover and housing for cracks. If leaks are suspected, use the dye-penetrant method to locate them.

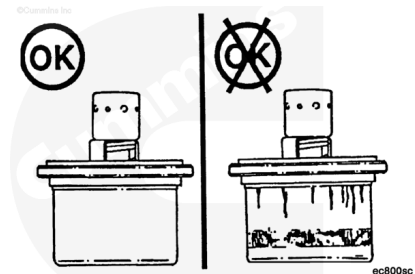
Check the cover and housing for corrosion.

Check the exterior of the element for scale-formed minerals in the cooling water. If there is evidence of scale, clean the exterior of the element using cleaning methods for the interior of the radiator.



07400262

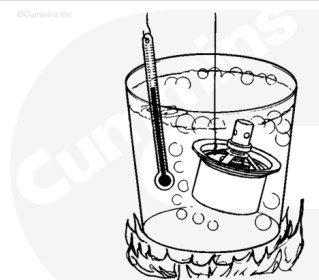
Inspect the thermostat for wear or damage. If the barrel of the thermostat is worn or fretted, the thermostat **must** be replaced.



ec800sc

Suspend the thermostat and a 100°C [212°F] thermometer in a container of water and heat the water to 100°C [212°F]. Do **not** let the thermostat or thermometer touch the sides of the container.

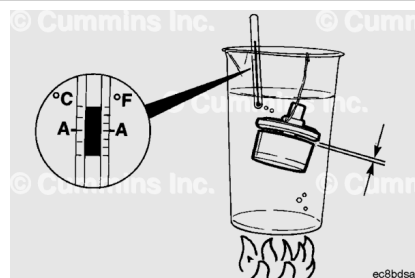
Allow the thermostat to soak in the 100°C [212°F] for 4 to 5 minutes.



ec200na

Continue to apply heat and check that the thermostat is fully open and the water is at the “full open” temperature.

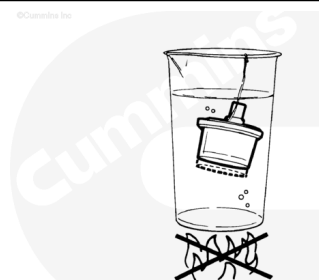
Thermostat Opening Distance			
	mm		in
100°C [212°F]	8	MIN	0.31



ec8bdsa

Remove the heat and check to see that the thermostat fully closes as the water cools to 85°C [185°F].

Measurements		
	celsius	fahrenheit
Fully Closed Thermostat	85	185



ec8bdsc

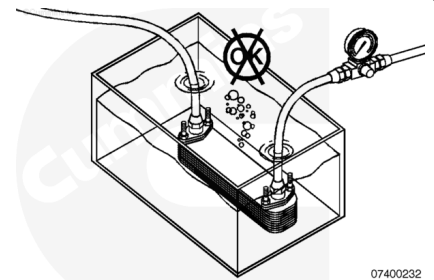
If the thermostat does **not** perform to specifications, it **must** be replaced.

Completely dry water from the thermostat prior to assembly.

Pressure Test

Apply air pressure of 415 kPa [60 psi] to the oil cooler element and check for leaks. Heat the water in the tank to 50°C [120°F] for more accurate test results.

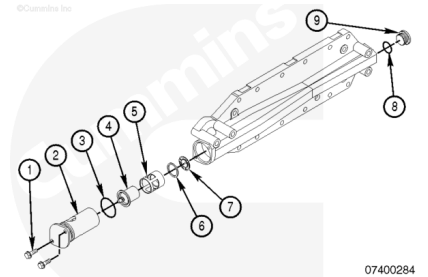
Discard element if leaks are present.



Assemble

⚠ CAUTION ⚠

The convex surface on the disk valve (7) must be positioned so that it faces the inside of the oil cooler housing and away from the thermostat. Engine damage can result if this is not installed properly.



Install the following components on the oil cooler cover:

- Plug (9)
- O-ring (8)
- Disk valve (7)
- Thermostat seal (6)
- Seal housing (5)
- Thermostat (4)
- O-ring (3)
- Cover (2)
- Capscrews (1).

Tighten the cover (2) capscrews.

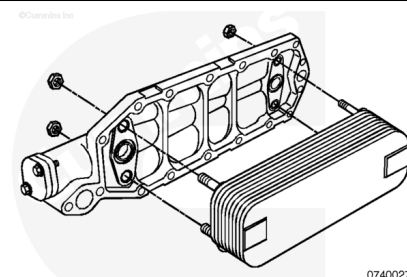
Torque Value: 31 n•m [23 ft-lb]

Tighten the oil cooler plug (9).

Torque Value: 17 n•m [151 in-lb]

⚠ CAUTION ⚠

If present, plastic shipping plugs must be removed from new oil cooler elements.



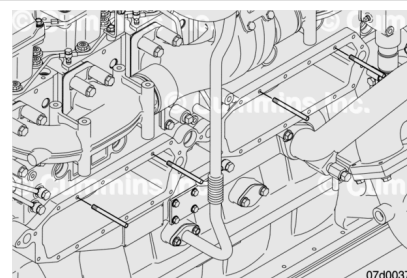
Install the oil cooler element to the oil cooler cover using new gaskets and o-rings.

Tighten the nuts.

Torque Value: 18 n•m [159 in-lb]

Install

Use guide studs as shown on the block to help with aligning the oil cooler.

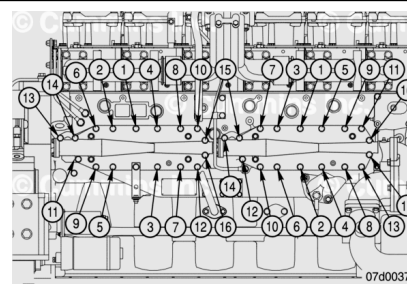


Install the new gasket, oil cooler assembly, and capscrews.

Follow the sequence for tightening the capscrews.

Tighten the capscews.

Torque Value: 66 n•m [49 ft-lb]



Finishing Steps

- Install the turbocharger oil supply and drain lines, if necessary. Refer to Procedure 010-033 (/qs3/pubsys2/xml/en/procedures/89/89-010-033.html) in Section 10.
- Install the water pump supply tube, if necessary. Refer to Procedure 008-062 (/qs3/pubsys2/xml/en/procedures/89/89-008-062-tr.html) in Section 8.

- Fill the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html) in Section 8.
- Check the level of the lubricating oil system. Add lubricating oil, if necessary. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/89/89-007-037.html) in Section 7.
- Operate the engine. Check for leaks.

Last Modified: 23-Oct-2019
